Hyungseok Kim

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Research Interests

Quantum Compilers, Quantum Error Correction, Quantum Programming, Quantum Algorithms.

Education

Yonsei University Seoul, Korea
M.S. in Electrical and Electronic Engineering Expected Feb. 2027
- Advisor: Prof. Won Woo Ro
- Thesis: "Efficient Lattice Surgery Compilation for Fault Tolerant Quantum Computing" — Based on [UR1].

Yonsei University Seoul, Korea
B.S. in Electrical and Electronic Engineering Feb. 2025
- Capstone Project: *HS-Table: A Table-Based Algorithm for Optimizing Quantum Circuits through Qubit Reuse* (Advisor: Prof. Won Woo Ro) — Expanded and published as [C3] in ISCA '25.

B.A. in Economics (Double Major) Feb. 2025

Research Positions

Yonsei University Seoul, Korea
- Graduate Researcher, *Embedded Systems and Computer Architecture Lab* (Advisor: Prof. Won Woo Ro) Mar. 2025 – Present
- Undergraduate Researcher, *Embedded Systems and Computer Architecture Lab* (Advisor: Prof. Won Woo Ro) Jun. 2024 – Feb. 2025

Publications (Google Scholar)

Under Review

[UR1] **Hyungseok Kim**, Changheon Lee, Seungjik Kim, Enhyeok Jang, Youngmin Kim, Seungwoo Choi, Hanbit Lee, Sungho Pyun, and Won Woo Ro. "SpiderLS: Leveraging Full ZX Reduction for Lattice Surgery Compilation." *arXiv preprint*, 2026. arXiv:2608.30228

Conference Papers

[C6] Sungho Pyun, Changheon Lee, **Hyungseok Kim**, Enhyeok Jang, Youngmin Kim, and Won Woo Ro. "DARTH: Lookahead-Driven Compilation Exploiting Transient Qubits for Distributed Quantum Computing." *The 59th IEEE/ACM International Symposium on Microarchitecture (MICRO '26)*, Athens, Greece, 2026. (Accepted for publication)

[C5] Changheon Lee, **Hyungseok Kim**, Seungwoo Choi, Youngmin Kim, and Won Woo Ro. "d'ArQ: A QOC Framework with Causality-Aware Grouping and Basis Selection." *2026 IEEE International Symposium on High Performance Computer Architecture (HPCA)*, Sydney, NSW, Australia, 2026. DOI:10.1109/HPCA68181.2026.11408534

[C4] Enhyeok Jang, **Hyungseok Kim**, Yongju Lee, Jaewon Kwon, Yipeng Huang, and Won Woo Ro. "Toward Scalable Gate-Level Parallelism on Trapped-Ion Processors with Racetrack Electrodes." *2026 IEEE International Symposium on High Performance Computer Architecture (HPCA)*, Sydney, NSW, Australia, 2026. DOI:10.1109/HPCA68181.2026.11408608 arXiv:2601.08930

[C3] **Hyungseok Kim**, Enhyeok Jang, Seungwoo Choi, Youngmin Kim, and Won Woo Ro. "QR-Map: A Map-Based Approach to Quantum Circuit Abstraction for Qubit Reuse Optimization." *The 52nd Annual International Symposium on Computer Architecture (ISCA '25)*, Tokyo, Japan, 2025. DOI:10.1145/3695053.3731020
- Merit Academic Paper Award, *Yonsei University Graduate School*, 2026.

[C2] Enhyeok Jang, Youngmin Kim, **Hyungseok Kim**, Seungwoo Choi, Yipeng Huang, and Won Woo Ro. "Qubit Movement-Optimized Program Generation on Zoned Neutral Atom Processors." *The 23rd ACM/IEEE International Symposium on Code Generation and Optimization (CGO '25)*, Las Vegas, NV, USA, 2025. DOI:10.1145/3696443.3708937 arXiv:2503.02272

[C1] Enhyeok Jang, Dongho Ha, Seungwoo Choi, Youngmin Kim, Jaewon Kwon, Yongju Lee, Sungwoo Ahn, **Hyungseok Kim**, and Won Woo Ro. "Recompiling QAOA Circuits on Various Rotational Directions." *International Conference on Parallel Architectures and Compilation Techniques (PACT '24)*, Long Beach, CA, USA, 2024. DOI:10.1145/3656019.3676899

Journal Papers

[J1] Youngmin Kim, Enhyeok Jang, **Hyungseok Kim**, Seungwoo Choi, Changheon Lee, Donghwi Kim, Woomin Kyoung, Kyujin Shin, and Won Woo Ro. "Distribution-Adaptive Dynamic Shot Optimization for Variational Quantum Algorithms." *Physical Review Research*, 7(4), 043253, 2025. DOI:10.1103/vwhk-22b8 arXiv:2412.17485

Workshop & Non-Archival Papers

[NC2] Junwoo You, **Hyungseok Kim**, Minchan Kim, and Won Woo Ro. "Profiling GPU-SSD I/O Behavior for Large Language Model Inference." *International Conference on AI for Science 2026 (AI4Sci Korea '26)*, Seoul, Korea, 2026. (Presented without proceedings)

[NC1] Changheon Lee, Enhyeok Jang, Youngmin Kim, **Hyungseok Kim**, Sungho Pyun, and Won Woo Ro. "TENET: A Pincer Movement with Backward Programs for Bypassing Relaxed Qubit Readouts." *The 25th International Conference on Electronics, Information, and Communication (ICEIC '26)*, Macau SAR, China, 2026. (Presented without proceedings)

[W2] **Hyungseok Kim**, Enhyeok Jang, Youngmin Kim, and Won Woo Ro. "Native Gate-Aware QAOA Ansatz." *The 1st International Workshop on Quantum Data and Machine Learning: Systems, Theory and Hardware (QDML '25)*, In conjunction with ICDE '25, Hong Kong SAR, China, 2025.

[W1] Enhyeok Jang, Youngmin Kim, **Hyungseok Kim**, Dongho Ha, Yongju Lee, Jaewon Kwon, Jun Woo You, Jiho Park, and Won Woo Ro. "A Dead Gate Elimination for Quantum Programs." *The 1st HPC/AI Integration with Quantum Computing Workshop (HAIQ '25)*, In conjunction with HPCA '25, Las Vegas, NV, USA, 2025.

Domestic Papers

[DC2] Sungho Pyun, Changheon Lee, **Hyungseok Kim**, Enhyeok Jang, Youngmin Kim, and Won Woo Ro. "Dynamic-Mapping Compilation Using Transient Qubits for Distributed Quantum Computing." *Summer Annual Conference of the Institute of Electronics and Information Engineers (IEIE)*, Jeju, Korea, 2026. (In Korean)

- Industry Interest Paper Award (LG Electronics), *The Institute of Electronics and Information Engineers (IEIE)*, 2026.

[DC1] Changheon Lee, Youngmin Kim, **Hyungseok Kim**, and Won Woo Ro. "How Basis Choice Shapes Performance in Quantum Optimal Control Framework." *Summer Annual Conference of the Institute of Electronics and Information Engineers (IEIE)*, Jeju, Korea, 2025.

[DM1] Youngmin Kim, Changheon Lee, **Hyungseok Kim**, and Won Woo Ro. "Research Trends and Prospects of Hybrid Computing-Based Variational Quantum Algorithms (VQAs)." *The Magazine of the Institute of Electronics and Information Engineers (IEIE)*, 52(9), 2025. (Magazine article, in Korean)

R&D Projects

University-Industry Foundation (UIF), Yonsei University Seoul, Korea

- Research Assistant, Optimizing GPU-SSD Integrated System Architecture for Large-Scale Artificial Intelligence Learning, *Samsung Electronics DS Division* Oct. 2025 – Sep. 2026
- Research Assistant, Research on Quantum Pulse Latency Reduction Through Quantum Optimal Control Framework, *National Research Foundation of Korea (NRF)* Jan. 2025 – Dec. 2025

- Research Assistant, Developing a High-Speed Quantum Circuit Simulator Supports Verification and Testing of Hyperscale Quantum Algorithms, *Hyundai NGV* Jun. 2024 – Oct. 2024

Professional Service

Reviewer

- Future Generation Computer Systems (FGCS) (SCIE Q1, JCR 2025) Oct. 2026
- The 59th IEEE/ACM International Symposium on Microarchitecture (MICRO '26) Artfact Evaluation Aug. 2026

Sub-Reviewer

- The 59th IEEE/ACM International Symposium on Microarchitecture (MICRO '26) May 2026
- The 53rd International Symposium on Computer Architecture (ISCA '26) Jan. 2026
- The 52nd International Symposium on Computer Architecture (ISCA '25) Jan. 2025

Student Volunteer

- The 58th IEEE/ACM International Symposium on Microarchitecture (MICRO '25) Oct. 2025

Laboratory Website Administrator

- Embedded Systems and Computer Architecture Lab, *Yonsei University* Mar. 2025 – Feb. 2027

Scholarships and Awards

Fellowships and Scholarships

- BK (Brain Korea) 21 Research Fellowship, *Yonsei University* 2025 – 2026
- Graduate Student Research Assistant (GSRA) Scholarship (New Student Excellence), *Yonsei University* 2025

Honors and Awards

- Industry Interest Paper Award (LG Electronics), *The Institute of Electronics and Information Engineers (IEIE)* 2026
- Merit Academic Paper Award, *Yonsei University Graduate School* 2026
- Student-Designed Major Competition, *Yonsei University* 2nd Place (with $360) 2024
- Academic Honors (Top 10%), *Yonsei University* 2024
- Micro-Learning Video Contest, *Yonsei University* 4th Place (with $380) 2023
- Academic Honors (Top 10%), *Yonsei University* 2022

Teaching Experience

Teaching Assistant, Yonsei University

- EEE 4473: Embedded System Lab. (Undergraduate) Spring 2025
- EEE 4610: Electrical and Electronic Engineering Capstone Design (Undergraduate) Spring 2025

Mentoring

- Research Mentor for Two Undergraduate Students, *Quantum Information Science Club Association (QISCA)* Nov. 2025 – Jan. 2026
- Academic Mentor for Two High School Students, *Dong Incheon High School* Mar. 2019 – Jun. 2019

Talks

Invited Talks

"SpiderLS: Leveraging Full ZX Reduction for Lattice Surgery Compilation"

- Group Meeting, *Topological Quantum Error Correction (TQEC) Design Automation* Sep. 2026

"QR-Map: A Map-Based Approach to Quantum Circuit Abstraction for Qubit Reuse Optimization"

- Journal Club, *Quantum Information Science Club Association (QISCA)* Sep. 2025

Contributed Talks

"Advanced LLM Systems: Reasoning Model"

- Large Language Models (LLMs) Seminar, *Yonsei University* Feb. 2026

"Advanced Techniques for Instruction Delivery and Speculation & Cross-Cutting Issues"

- Advanced Computer Architecture Seminar, *Yonsei University* Feb. 2025

"Understanding Interrupt Signals in Linux Kernel"
- Advanced Operating Systems Seminar, *Yonsei University* Jan. 2025

Skill Set

Technical Qualifications
- Adobe Certified Professional (ACP) in Digital Video Using Adobe Premiere Pro, *Adobe Inc.* 2024
- General Computer User, Level-I (Highest), *Korea Chamber of Commerce and Industry* 2020

Programming Languages: C/C++ (OpenMP, CUDA), Python (Qiskit), JavaScript, MATLAB, Assembly Languages (RISC-V, ARM)

Other Languages and Tools: Verilog, LaTeX, Markdown, HTML, YAML, CSS, Jekyll, Git

Languages

Working Proficiency: Korean (Native), English (CEFR C1, TOEFL iBT 5.0/6.0)

In Progress: Spanish, Chinese, Japanese

Extracurricular Activities

Professional Affiliations
- Graduate Student Member, *Institute of Electrical and Electronics Engineers (IEEE)* 2025 – Present
- Trainee, *Automotive System IC Fusion Human Resource Research Center (ASRC)* 2025 – 2026
- Trainee, *Korea Collaborative & High-tech Initiative for Prospective Semiconductor research (K-CHIPS)* 2026

Student Activities
- Participant, IonQ Research Mentoring Program, *IonQ* 2025
- Participant, Language Exchange Program, *Yonsei Korean Language Institute (KLI)* 2023 – 2024
  - 5 Quarters: 4 English–Korean, 1 Spanish–Korean
- Participant, Virtual Pen Pal Program, *University of South Florida* 2023
- Participant, Waseda-Yonsei Campus & Cultural Exchange, *Waseda University* 2019

Military Service
- Personnel Administrator / Sergeant (E-5), *142nd Intelligence Battalion, II Corps, Republic of Korea Army* 2020 – 2022

Miscellaneous

- Technical Blog: fluorite42.github.io/categories
  - "An Introduction to Quantum Error Correction" (5-part series) 2025
  - "A Guide to Quantum Gates" (2-part series) 2025
- Independent Global Traveler: Backpacked 48 Cities across 30 Countries 2023 – Present
- Erdős Number: 4